Project title: Drone Detection using YOLO11

- **Introduction**:
  - UAVs (drones) are widely used in surveillance, delivery, mapping and disaster monitoring, but their misuse near airports, defense areas and public events is a growing security threat.
  - This project aims to build a reliable visual UAV detector that works on **small, distant drones** across **different backgrounds** (open sky, urban, forest).
  - The work builds on an existing pipeline (YOLOv9 + EfficientNet-B3) and aims to improve it using a more recent, lighter single-stage detector.
- **Motivation**:
  - Security agencies need a system that can **quickly and accurately identify UAVs** to enable timely action against potential threats.
  - Drones are hard to detect because they often appear as **just a few pixels** in the image and can look similar to birds or kites.
  - A **lighter, single-stage detector** is preferred over a heavier two-stage pipeline for faster inference and future edge deployment.

Name: Gajendra Gangwar | Roll No: 2301081 | Supervisor: Dr. Rakesh Matam

Existing work / Literature review:

- **Existing work:**
  - The existing in-house pipeline used **YOLOv9** to detect drone regions and **EfficientNet-B3** to re-classify the detections, reducing false positives.
  - The model was trained for **20 epochs** on the same drone/non-drone dataset used in this project.
- **Performance of the existing model:**

Environment-wise confidence scores:

Environment	Confidence Score
Bright / Clear sky	~1.00
Night / Low-light	0.85 – 0.98
Cluttered	0.70 – 0.80

Distance-wise confidence scores:

Distance	Confidence Score
Close-range	0.93 – 1.00
Mid-range	0.90 – 0.96
Long-range (very far)	0.70 – 0.78

- **Gaps in the existing work:**
  - Accuracy **drops sharply on very-far / tiny drones** (confidence as low as 0.70).
  - Performance **degrades in cluttered backgrounds** (forests, urban) with confidence only 0.70–0.80.

Work Done till date:

- **Proposed approach:**
  - YOLO11-nano (~2.6M parameters) was selected for its dedicated **C2PSA attention block** for small-target detection, an **anchor-free head with DFL loss** for tighter box localisation. It was trained for **500 epochs** on the drone dataset.
  - Built **two custom validation splits** for detailed evaluation:
    - **Distance-wise:** Close / Mid / Long / Multi-range.
    - **Environment-wise:** Bright clear / Cluttered / Low-light.
  - **Key hyperparameters:** image 320×320, batch 2, AdamW optimiser, patience 30 epochs.

- **Results — overall comparison with existing model:**

Metric	YOLOv9 + EffNet-B3	YOLO11-nano
Precision	0.9383	0.9622
Recall	0.8953	0.8662
mAP@50	0.9307	0.9305
mAP@50–95	0.6214	0.6727
Box loss	0.9760	0.7614
Cls loss	0.6431	0.3774
DFL loss	1.2123	0.8826

- **Key takeaways:**

YOLO11-nano performs significantly better in the two hardest conditions :
Long-range drones: mAP@50 = **0.9066** (existing model confidence was only 0.70–0.78)
Cluttered backgrounds: mAP@50 = **0.9134** (existing model confidence was only 0.70–0.80)

- **Results:**

Distance-wise results:

Distance	Precision	Recall	mAP@50
Close-range	0.9533	0.8852	0.9534
Mid-range	0.9843	0.9690	0.9889
Long-range	0.9496	0.8479	0.9066
Multi-range	0.9579	0.8810	0.9273

Environment-wise results:

Environment	Precision	Recall	mAP@50
Bright / Clear sky	0.9664	0.8943	0.9486
Night / Low-light	0.9695	0.8840	0.9600
Cluttered	0.9527	0.8425	0.9134